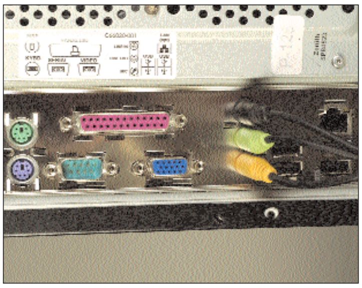
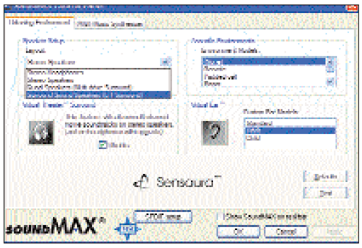

Final Subwoofer assembly


The soundcard all hooked up

control panel or the sound card drivers. All sound cards come with a setup Wizard or calibration Wizard—click on it. Here, too, select "5.1": some software will intelligently detect the speaker connects and show 5.1. These utilities usually play sounds from each speaker while indicating which speaker is playing by highlighting them on the screen. This is to ensure that the channels are right. If the speakers are playing when they should, breathe a sigh of relief—you're nearly through!

If a sound channel has got mixed up, you may have either connected the speakers to the back of the woofer/amp incorrectly, or the outputs at the sound card end have been plugged in wrong. The most common problem is of left and right or front and rear being inverted; simply swap the connectors. If the cables and the sound card sockets aren't colour coded, and if there's no sound from, for example, the centre speaker, you can use trial and error to get it right.

7. Another important tweak is the subwoofer crossover. In your sound card drivers, this will be shown as a slider bar. Generally, if set at a value like 120 Hz (typically, the crossover allows a range from 0 to 200 Hz), it means that any sound intercepted by the decoder unit below 120 Hz will be redirected to the woofer unit. The typical crossover frequency should be between 80 Hz and 120 Hz; anything higher than this can be handled by the mid range (squawker) drivers. Experiment and see what sounds best for you. The best sound may even vary from song to song, but there generally is an optimal setting.



Changing the Speaker Options in your soundcard panel

8. Slip a DVD movie into your drive. Start her up! You now need to calibrate the volume of the speakers. A good way to start is to set the overall (or "master") volume to between 50 and 75 per cent, while playing around with the individual speaker volumes. The woofer volume should be kept at about 60 per cent; this is a comfortable volume that doesn't distort, and at the same time, doesn't intrude on the other sounds. Usually, the centre speaker is a bit more powerful than the other satellites, so it can be set at a slightly higher volume level than them without distortion. The centre speaker should, as a rule always be more powerful, because this is the channel that produces the vocals in a 5.1 movie.

9. Sit back on your settee, grab the popcorn (which you dropped in frustration earlier), and chill out! ■

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Jargon Buster

LFE: An acronym for "Low Frequency Effects". The very term "sub" in the term "subwoofer", means meaning sound below 50 Hz. A sub is typically capable of producing sound in the range of 35 Hz to 100 Hz. Some subs can go higher, though this is not really needed. A sub that can go lower than 35 Hz is good indeed, and the lower a sub can go, the more ideal it is for producing deep rumbles. Subs that go really low are also costlier.

Drivers: The driver is what produces the sound from your speakers. It consists of the enclosure (typically metal), a permanent magnet, and the diaphragm (often of paper). Speaker drivers are of three types, distinguished largely on the frequency spectrum they are built to reproduce. Tweeters are the smallest of the speaker drivers and can produce the highest-frequency sounds, like the tinkling of breaking glass. A good-quality tweeter can reproduce sounds up to 25,000 Hz. Squawkers are the mid-range drivers. They are typically 3- to 5-inch drivers with the ability to produce a wide spectrum of sound between the sub and the tweeters.

The subwoofer is the low-frequency sound producer, and is responsible for all the thuds and thumps. They are capable of playing sounds below 150 Hz. A sub also has the biggest driver, sizes range from 6.5 inches to 12 inches in diameter.

SNR: The Signal to Noise Ratio is a very important consideration for audiophiles and the discerning user. Very simply put, every sound channel carries, along with the actual audio content, a certain amount of "signal interference" or stray noise. This causes distortion in sound, and it is particularly noticeable at higher volumes, mainly because of the unit design imperfections. SNR is measured as a ratio. A typical good value for a speaker is 85 dB, with some higher end 5.1 sets having an SNR as high as 110dB. Speakers with an SNR of below 65 dB should be avoided at all costs!